GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

242439.1.pdf *Geochronology Record 1918*

242439.1: granodiorite gneiss, Mt Marvel quarry

(Yilgarn Craton granites; Youanmi Terrane, Yilgarn Craton)

SS Romano, Y Lu, IOH Fielding, MTD Wingate and TJ Ivanic

Location and sampling

PERENJORI SH50-06, PERENJORI 2139 Zone 50, MGA 411691E 6780029N
Warox Site TJIROT190059 Sampled on 15 May 2019

This sample was collected from the southern side of the Mount Marvel quarry, about 10.6 km east of Pintharuka, 3.6 km northwest of Pinjarra Hill, and 1.7 km west of the Morawa Yalgoo Road.

Geological context

The unit sampled is a granodiorite gneiss of Yilgarn Craton granites (Fig. 1). A metamonzogranite of the Bunnawarra Suite, Thundelarra Supersuite, sampled about 32 km to the east, yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al. 2020). Although the 3010–2930 Ma Thundelarra Supersuite was previously considered the oldest basement unit in the Youanmi Terrane (Ivanic et al., 2022), the igneous crystallization age of this sample discussed below suggests existence of even older basement up to at least c. 3275 Ma in the Youanmi Terrane.

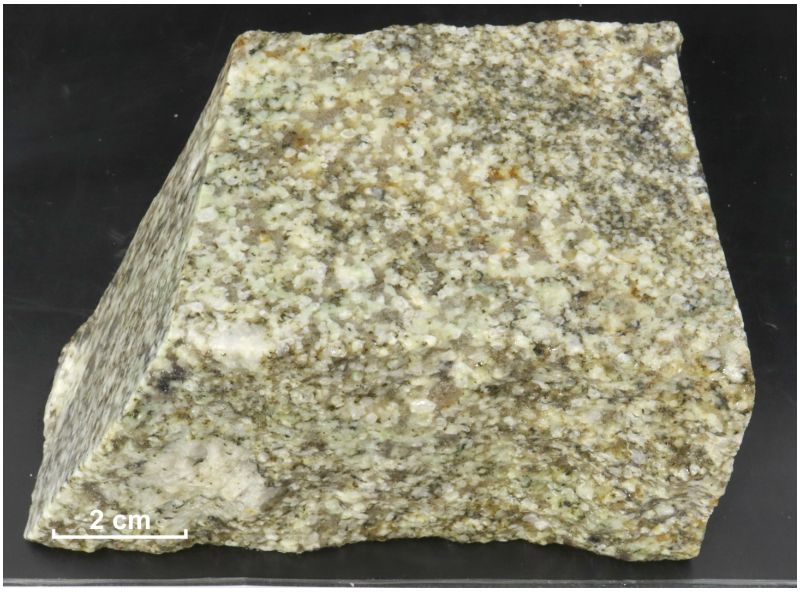

Figure 1. Hand specimen image for sample 242439: granodiorite gneiss, Mount Marvel quarry

Petrographic description

The sample is a fine- to medium-grained, equigranular biotite granodiorite gneiss, consisting of about 55% plagioclase, 25% quartz, 15% K-feldspar, 5% biotite and accessory magnetite, apatite and zircon. Plagioclase occurs as subhedral, mainly equant to slightly tabular grains to 2.5 mm in size with most grains altered by sericite and minor muscovite. Quartz occurs as anhedral grains up to 1 mm in size, forming irregular aggregates between larger feldspar grains, or as recrystallized polygonal grains to 0.1 mm in diameter throughout the sample. K-feldspar is perthitic, occurs as anhedral grains to 2 mm in diameter, and is often fringed by recrystallized polygonal grains of mainly K-feldspar. Some relict K-feldspar contains inclusions of plagioclase and quartz. Biotite is brown and occurs as subhedral laths to 0.5 mm in length within small aggregates of either decussate or well-aligned grains, mostly around the margins of feldspar. Biotite is also a minor component of the recrystallized aggregates. Magnetite occurs as disseminated, anhedral and subhedral grains within small aggregates of biotite. Apatite occurs as euhedral grains to 0.1 mm in size. Partial recovery involving grain boundary adjustment in quartz and minor fine-grained recrystallization indicate a moderate metamorphic overprint.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and subhedral to euhedral. The crystals are up to 250 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and some crystals appear to contain older or metamict cores. A CL image of representative zircons is shown in Figure 2.

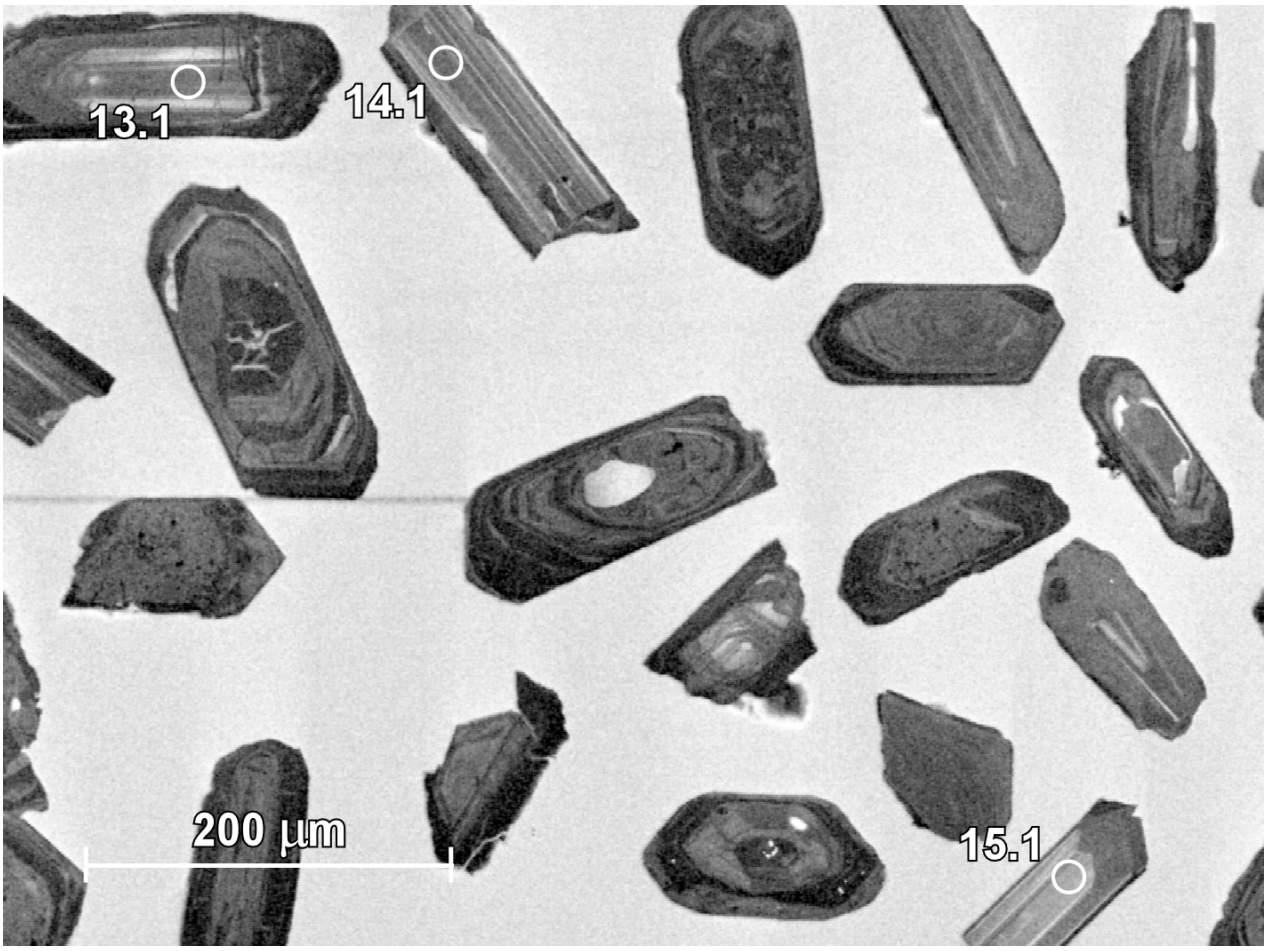

Figure 2. Cathodoluminescence image of representative zircons from sample 242439: granodiorite gneiss, Mount Marvel quarry. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 10–11 February 2022, using SHRIMP-B. Fourteen analyses of the M257 standard were obtained during the session, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.14% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 2). One analysis is >8% discordant. The date obtained from this analysis (Group D, Table 1) is unreliable and considered not to be geologically significant. The remaining 17 analyses can be divided into four groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 11 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 3275 ± 5 Ma (MSWD = 2.1).

Group X comprises one analysis of a zircon core (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 3506 ± 4 Ma (1σ).

Group P comprises three analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3260–3250 Ma.

Group Z comprises two analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of c. 1492 and 1491 Ma.

The date of 3275 ± 5 Ma for the 11 analyses in Group I is interpreted as the igneous crystallization age of the granodiorite protolith.

The date of 3506 ± 4 Ma for the single analysis in Group X is interpreted as the age of an inherited component. The dates of 3260–3250 Ma for the three analyses in Group P are interpreted to reflect minor ancient loss of radiogenic Pb. The dates of c. 1492 and 1491 Ma for the two analyses in Group Z are interpreted as the ages of contaminants introduced during sample processing.

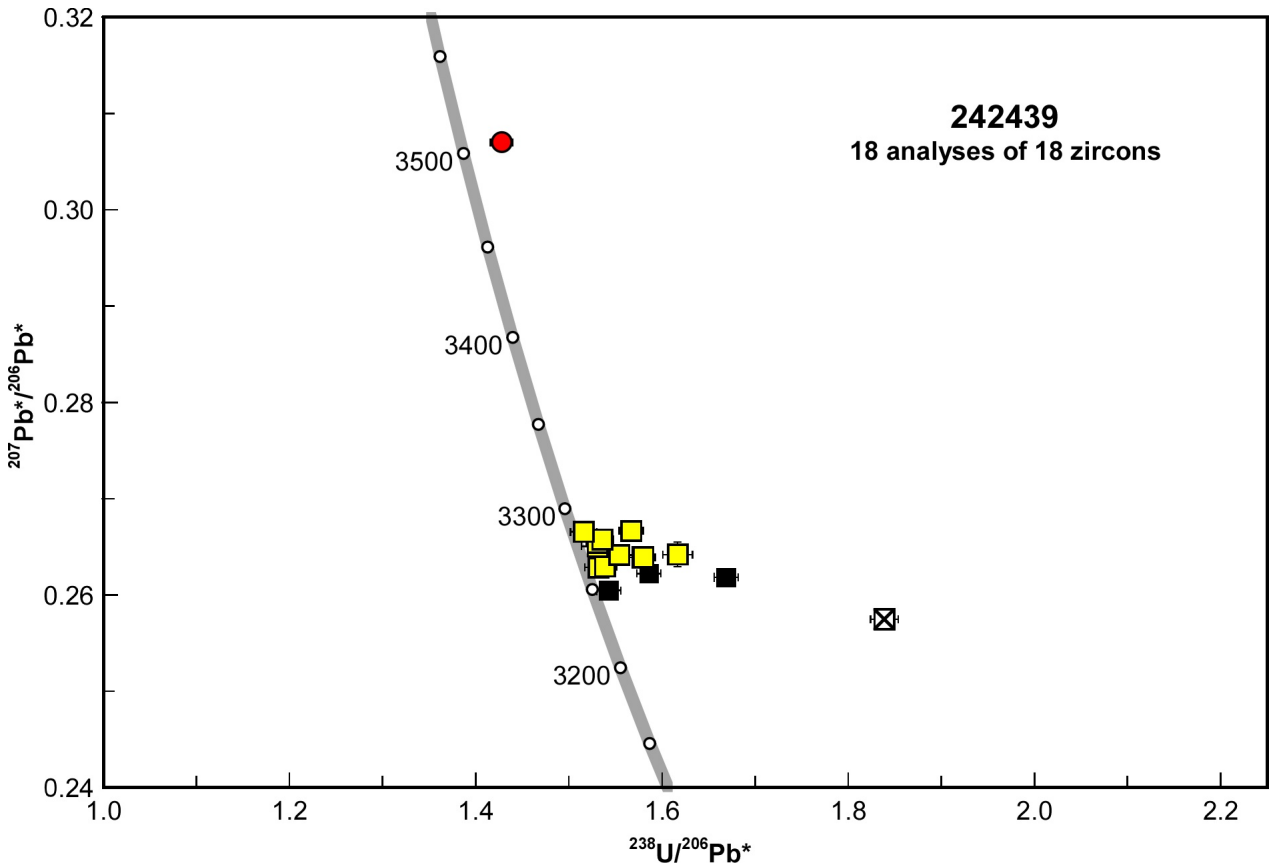

Figure 3. **U–Pb analytical data for zircons from sample 242439: granodiorite gneiss, Mount Marvel quarry. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon core); black squares indicate Group P (radiogenic-Pb loss); crossed square indicates Group D (discordance >8%). Two analyses in Group Z are not shown**

Links

Table 1. Ion microprobe analytical results for zircons from sample 242439: granodiorite gneiss, Mount Marvel quarry: 242439.1.Table1.xls
SHRIMP spot images: 242439_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

References

Ivanic, TJ, Wingate, MTD, Lowrey, JR and Lu, Y 2022, Formation of the Yilgarn protocraton by rift-related magmatism from 3.01 to 2.92 Ga: Geological Survey of Western Australia, Report 232, 34p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Wingate, MTD, Lu, Y and Zibra, I 2020, 219311: metamonzogranite, North Paradise Well; Geochronology Record 1680: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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The Government of Western Australia acknowledges Aboriginal people as the traditional custodians throughout Western Australia and their enduring connection to the land, waterways and community. We pay our respects to Elders past and present.

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